FIT2179 Data Visualisation

Course Notes

CHOROPLETH MAPS

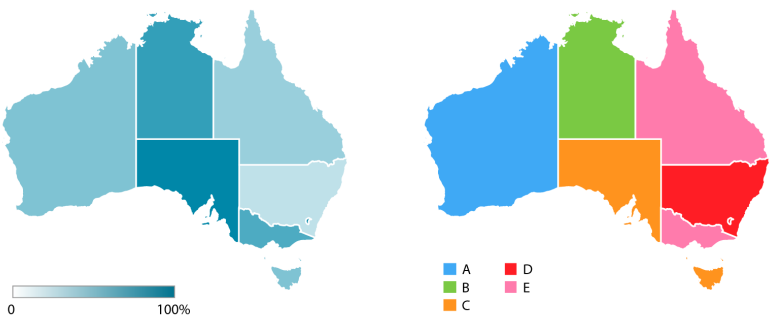
12 September 2026

CHOROPLETH MAPS

TERMINOLOGY

Choropleth combines two Greek words: ch*oros* means area or region, and *pleth* or *plethos* means multitude, as in plethora.

A choropleth map uses geographic area marks with colour channels. It can show a single categorical attribute with the hue channel, or a single quantitative attribute with the brightness channel.



COLOUR FOR QUANTITATIVE CHOROPLETH MAPS

- Primarily brightness changes: the greater the value, the darker the colour.
- While brightness is the main channel, it is common for choropleth maps to also vary saturation and hue within limits to increase the number of distinguishable colours, and also for aesthetic reasons.
- Diverging distributions combine two hues (for an example, see *Granularity of Areas*).
- Varying saturation is possible, but should not be the main channel.

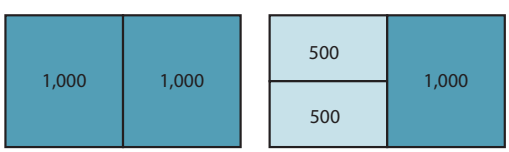
An exception applies for quantitative choropleth maps with a dark background, where greater values can be shown with brighter values.

STANDARDISING ATTRIBUTE VALUES

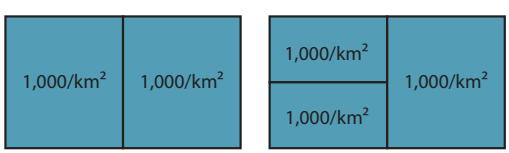
It is often important to standardise raw quantitative values to compensate for the differing size of areas in a choropleth map. Showing raw values without standardisation in choropleth maps is a common error.

To see why standardisation matters, consider a choropleth map with only two areas. Assume both areas are the same size (1 km²) and have the same number of residents (1,000). The left-hand schema below illustrates the resulting map when showing raw values.

Now suppose the left area is split into two areas of 0.5 km² each, with 500 people living in each. Because the raw values are smaller, the new areas are brighter (below right).

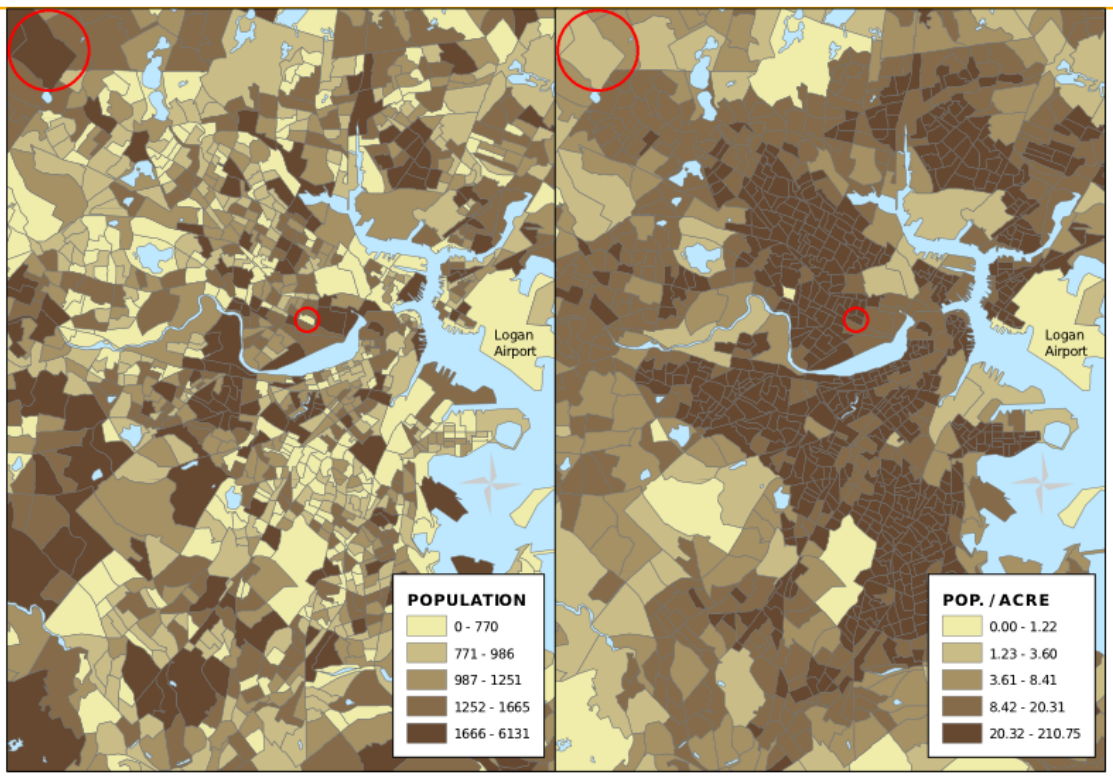


Yet there is no reason for the map to show a different pattern because no resident has moved. If the number of residents in each area is divided by its size, the two schematic maps below show population density correctly, both before and after the subdivision.



Total Population of 2000 Census Block Groups

Population Density of 2000 Census Block Groups



This is an example illustrating the need for standardising raw values. The two choropleth maps show the urban area of Boston in the USA. The purpose of the maps is to show the distribution of population. The viewer should be able to identify areas with dense and sparse populations.

The map on the left shows raw population counts. The number of people counted in each area is directly mapped to brightness.

This map is wrong and misleading because it does not compensate for the differing size of the areas. This map is not useful because it cannot show the pattern of population distribution.

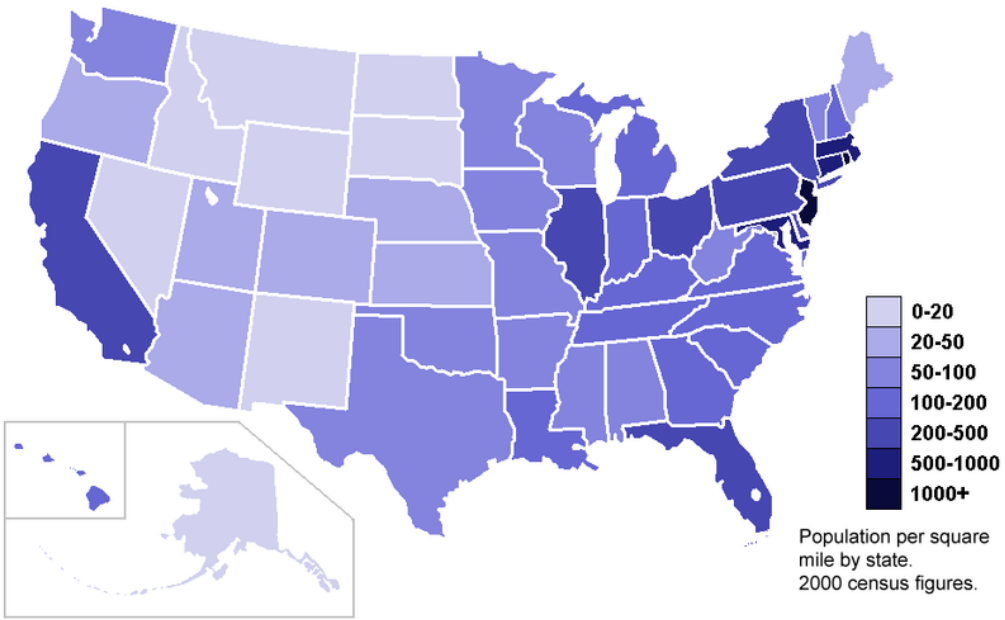
The map on the right shows population density: the number of people divided by the size of each area. This is the correct map, as it clearly shows areas with dense and sparse populations. Clusters of dark brown colour with high population density are easy to identify.

More generally, standardisation means dividing a raw count by something that makes values comparable across regions. Standardisation by surface area is only one choice. Counts are often divided by population (giving a rate per capita) or by a total (giving a percentage). Which denominator to use depends on what the map should show.

Maps from Slocum et al. 2022, Thematic Cartography and Geovisualization, ed. 4.

CLASSIFICATION OF ATTRIBUTE VALUES

Many quantitative choropleth maps show classified values. Classification means that each shown value is first assigned to a class. The map below uses seven classes, as shown by the legend. Each class has an upper and a lower boundary.



PROJECTIONS FOR CHOROPLETH MAPS

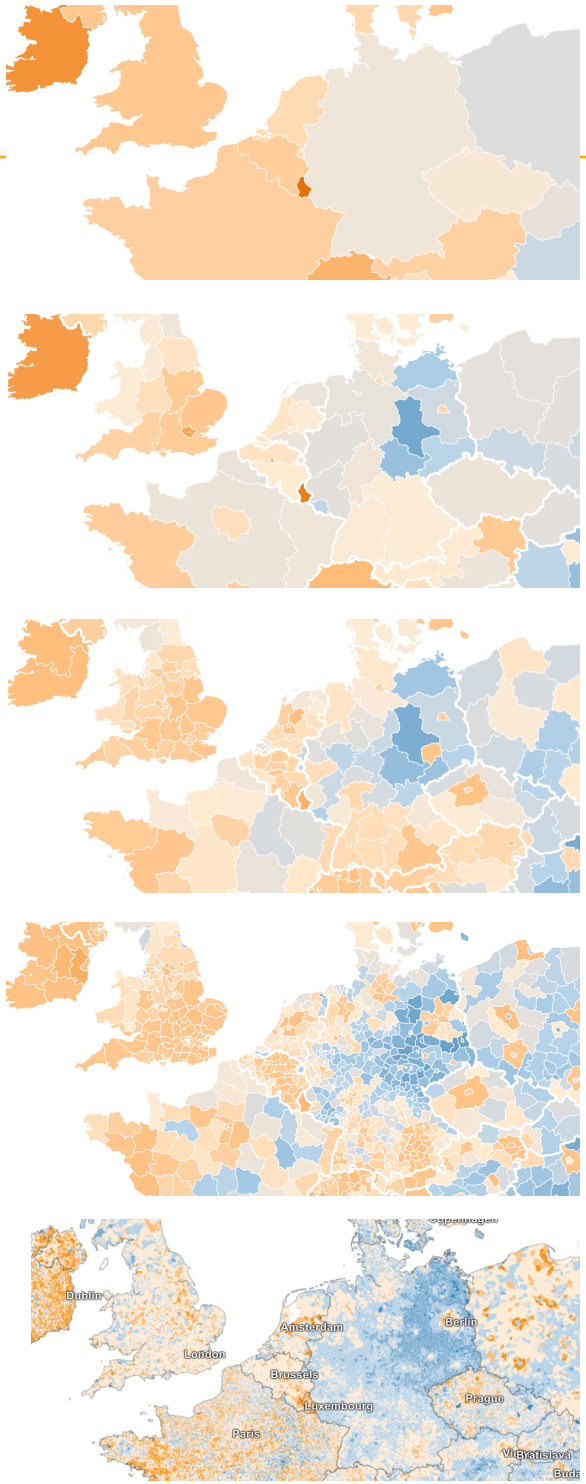
Choropleth maps showing a large geographic area, such as a large country, a continent, or the entire world, should use a projection that does not distort area. The Mercator projection or any other projection that adds considerable areal distortion is a poor choice for a choropleth map because the enlarged areas visually dominate the map.

GRANULARITY OF AREAS

Sometimes data is available at different geographic granularities, and the map author gets to choose granularity or level of detail. The series of maps on the right show population growth.

Population growth, annually

- -3%
- -1%
- 0%
- +1%
- +3%



The first map shows European countries. The map is unable to show interesting patterns because there are only a small number of areas, and the areas have vastly different dimensions. The following maps increase the granularity and are able to show more interesting patterns. For example, the fourth map shows that the eastern part of Germany lost population, while the southern part of Germany increased in population. The last map shows data at the municipality level, where patterns across larger areas are difficult to identify, but the changes in population of individual cities are more clearly visible.

Hence, the purpose of a map will determine what level of granularity it should show.

Source:
https://www.datawrapper.de/blog/weekly-chart-europegrowth